CRASH Competition Team 2 Statement of Work

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Project Description:

The goal of the CRASH (Controlled Recovery and Safe Handling) Competition is to introduce students to the engineering principles behind launching, protecting, and safely recovering fragile payloads. In this competition, eggs and water are used as representative payloads that require careful design to prevent damage during flight and landing. Team 2 has been broken down into five subsystems: the egg payload, water payload, propulsion and recovery, avionics, and the airframe. The airframe subsystem includes the vehicle body and motor, ensuring a successful flight. The avionics system collects flight data, recording altitude and GPS information. The payload system focuses on impact and vibration protection, while propulsion and recovery ensure a controlled and safe ascent and descent.

1) Long Term Objectives and Payoff

The long-term objective for the competition and project is to provide students with hands-on experience in rocketry design, payload protection, and recovery, allowing them to apply theoretical concepts to industry scenarios in the future. Team 2’s objective is to successfully launch and recover the rocket while ensuring the safety of the fragile payloads it carries.

2) Current Semester Objectives

The current semester objectives focus on the complete manufacturing, testing, and integration of all the subsystems. Each subsystem will be fabricated and tested separately to ensure subsystem functionality, and then integrating the vehicle as a whole. Measuring parameters include ground testing of the recovery system, verifying avionics data collection for altitude, GPS, and camera recording, and testing payload capsules with drop tests. The main objective in integration is ensuring all systems fit inside the rocket and do not exceed our weight maximum for the goal altitude. These objectives ensure the project will be flight ready and safe for launch at the CRASH competition.

3) Technical Approach

The technical approach begins with defining and procuring all required materials for each subsystem, documented in a project Bill of Materials to ensure compatibility and cost control. Once materials are obtained, fabrication and validation of all subsystems will proceed.

Aerostructural components, including the body tube and internal structures, will be machined or additively manufactured using UCF facilities. These components require precise tolerances and alignment with subsystem interfaces, and design or manufacturing issues will be addressed through iterative refinement.

Rocket bays will be fabricated and structurally tested to verify safety, functionality, durability, and reusability. Test results will guide any necessary redesign or re-manufacturing.

The recovery system will be tested under representative loading conditions to verify shock cord strength and parachute deployment. Controlled low-altitude drop tests will be used to evaluate descent behavior and landing performance.

Avionics and electrical systems will be tested to ensure reliable operation under launch conditions, including sensor performance and data acquisition. After subsystem validation, the rocket will be fully assembled, undergo final inspections, and be prepared for launch, with flight data collected for post-flight analysis.

4) Project Plan with Deliverables and Dates

To start the project the plan begins with manufacturing all the systems, which is estimated to take about 7-8 weeks. After the vehicle components have been manufactured, a testing and validation phase shall begin, with 2-3 weeks allocated for this phase. Once the vehicle has been validated, the time scheduled to assemble the final vehicle build is another 2-3 weeks. This plan timing allows the team a comfortable window between the initial and final launch attempt, with a fallback plan in place should certain aspects not pan out accordingly for the first launch date.

Manufacturing	1/12/26-2/27/26
Material and parts order	1/12/26-1/27/26
Material and parts acquisition	1/27/26-1/31/26
Body Tube Machining	2/2/26-2/3/26
Fin, Bulkhead, & Centering Ring Fabrication	2/2/26-2/5/26
Nosecone Fabrication	2/2/26-2/11/26
Electronics Bay Fabrication & Assembly	2/2/26-2/27/26
Egg Payload Bay Fabrication	2/2/26-2/27/26
Water Payload Bay Fabrication	2/2/26-2/27/26
Testing and Validation	3/2/26-3/15/26
Payload safety testing	3/2/26-3/13/26
Ejection Charge Testing	3/2/26-3/13/26
Shock Cord Load Testing	3/2/26-3/13/26
Payload Load/Unload Time Testing	3/2/26-3/13/26
Avionics Testing	3/2/26-3/13/26
Placeholder for last minute manufacturing tweaks	3/13/26-3/15/26
Placeholder for any additional testing needed	3/13/26-3/15/26
Final Build	3/7/26-3/21/26
Project completion goal	3/7/26-3/21/26
Inner Tube & Centering Ring Fixing to Body Tube	3/9/26-3/13/26
Fin Fixing to Body Tube	3/9/26-3/13/26
Nose Cone Fixing to Body Tube	3/9/26-3/13/26
Launch	3/21/26 and 4/18/26
Initial launch attempt	3/21/26
Final launch attempt	4/18/26
Fallback Launch Attempt Plan	3/23/26-4/18/26
Additional Payload safety testing	3/23/26-4/10/26
Additional Ejection Charge Testing	3/23/26-4/10/26
Additional Shock Cord Load Testing	3/23/26-4/10/26
Additional Payload Load/Unload Time Testing	3/23/26-4/10/26
Additional Avionics Testing	3/23/26-4/10/26
Placeholder for last minute manufacturing tweaks	4/10/26-4/12/26
Placeholder for any additional testing needed	4/10/26-4/12/26
Final Build	4/13/26-4/17/26